

Bayesian Inference

Principles, Methods, and Applications

Graduate Seminar in Quantitative Methods

2026

Outline

- 1 Motivation
- 2 Bayes' Theorem
- 3 Prior, Likelihood, and Posterior
- 4 Conjugate Priors and the Beta–Binomial Model
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- 6 Hierarchical Models
- 7 Computational Methods
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Two Paradigms of Statistical Inference

Frequentist

- Parameters θ are *fixed* but unknown
- Probability = long-run frequency
- Inference via repeated sampling
- Results: p -values, confidence intervals
- No prior information encoded for θ

Bayesian

- Parameters θ are *random variables*
- Probability = degree of belief
- Inference by updating beliefs with data
- Results: posterior distributions, credible intervals
- Incorporates prior knowledge about θ

Key Question

Given data $\mathcal{D}$, what can we say about the unknown parameter θ ?

Why Bayesian Inference?

Compelling Features

- **Coherent uncertainty:** the posterior $p(\theta \mid \mathcal{D})$ is a complete probability statement about θ
- **Sequential updating:** yesterday's posterior is today's prior — ideal for streaming data
- **Natural regularisation:** the prior acts as a soft constraint, reducing overfitting
- **Decision-theoretic framework:** Bayes risk $= \mathbb{E}_{\theta, \mathcal{D}}[\ell(\hat{\theta}, \theta)]$ is minimised by the Bayes estimator
- **Hierarchical modelling:** borrows strength across groups via shared hyperpriors

Classic Applications

Medical diagnosis, spam filtering, A/B testing, astrophysics (LIGO gravitational waves), NLP language models, reinforcement learning from human feedback

From Conditional Probability to Bayes' Theorem

Definition of conditional probability:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad \mathbb{P}(B) > 0$$

Joint probability written two ways:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$$

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$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \mathbb{P}(A)}{\mathbb{P}(B)}$$

Law of Total Probability supplies the denominator:

$$\mathbb{P}(B) = \sum_i \mathbb{P}(B \mid A_i) \mathbb{P}(A_i)$$

Bayes' Theorem for Statistical Inference

Replace event $A \rightarrow \theta$ (parameter), event $B \rightarrow \mathcal{D}$ (data):

Bayes' Theorem

$$\underbrace{p(\theta | \mathcal{D})}_{\text{posterior}} = \frac{\overbrace{p(\mathcal{D} | \theta)}^{\text{likelihood}} \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(\mathcal{D})}_{\text{marginal likelihood}}}$$

The marginal likelihood (“evidence”) normalises the posterior:

$$p(\mathcal{D}) = \int_{\Theta} p(\mathcal{D} | \theta) p(\theta) d\theta$$

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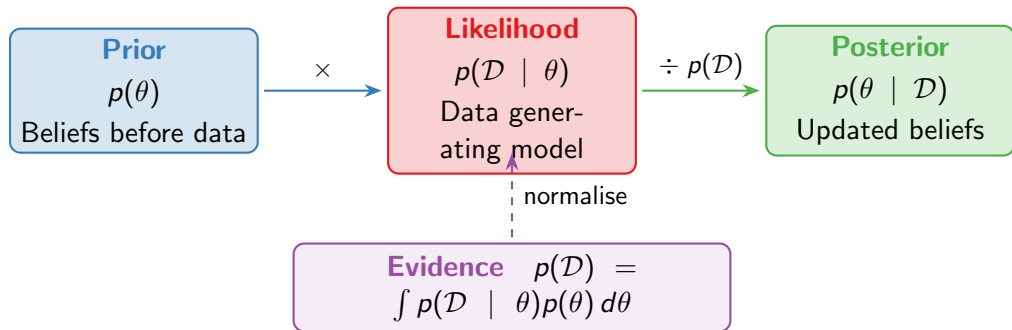
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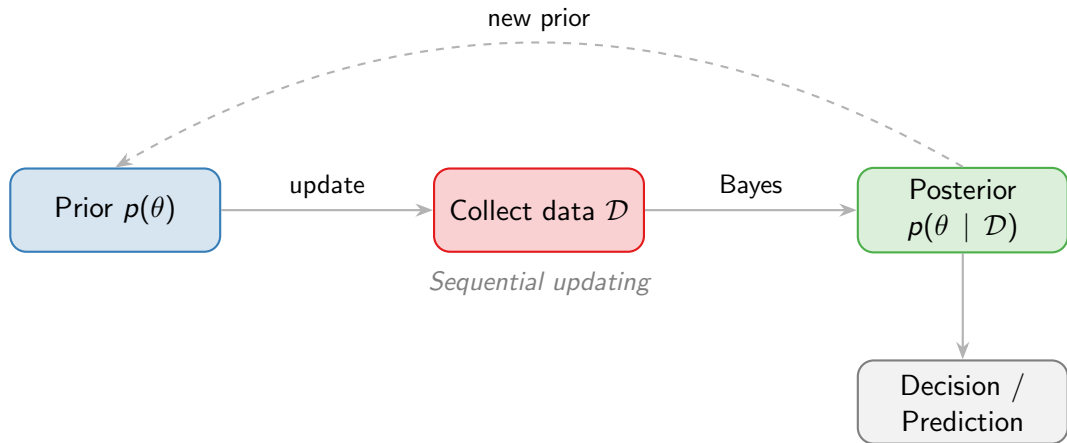
Proportionality Shortcut

Since $p(\mathcal{D})$ is constant in θ :

The Three Components



The Bayesian Update Cycle



- **Today's posterior is tomorrow's prior** — the cycle repeats as data arrive
- The final answer is the full posterior distribution, not a single number

The Prior Distribution

The prior $p(\theta)$ encodes knowledge *before* observing data.

Types of priors

- **Informative:** strong domain knowledge
 $\theta \sim \mathcal{N}(0.5, 0.1^2)$
- **Weakly informative:** regularising
 $\theta \sim \mathcal{N}(0, 1^2)$
- **Non-informative / reference:**
 $p(\theta) \propto 1$ (flat), Jeffreys:
 $p(\theta) \propto \sqrt{I(\theta)}$
- **Conjugate:** posterior in same family (next section)

Jeffreys Prior

Invariant to reparametrisation:

$$p_J(\theta) \propto \sqrt{\det \mathcal{I}(\theta)}$$

Fisher information matrix:

$$\mathcal{I}(\theta) = -\mathbb{E} \left[\frac{\partial^2 \log p(\mathcal{D} \mid \theta)}{\partial \theta \partial \theta^\top} \right]$$

For $\theta \in (0, 1)$, Binomial:

$$p_J(\theta) \propto \theta^{-1/2} (1 - \theta)^{-1/2} \Rightarrow \text{Beta}(\tfrac{1}{2}, \tfrac{1}{2})$$

The Likelihood Function

Definition

Given data $\mathcal{D} = \{x_1, \dots, x_n\}$ and model $p(x | \theta)$:

$$\mathcal{L}(\theta; \mathcal{D}) = p(\mathcal{D} | \theta) = \prod_{i=1}^n p(x_i | \theta) \quad (\text{i.i.d. assumption})$$

- The likelihood is a **function of** θ , not a probability over θ
- It need not integrate to 1 over θ
- Work with the log-likelihood for numerical stability:

$$\ell(\theta) = \log \mathcal{L}(\theta; \mathcal{D}) = \sum_{i=1}^n \log p(x_i | \theta)$$

MLE vs MAP

MLE: $\hat{\theta} = \arg \max_{\theta} \ell(\theta)$

Conjugate Prior Families

A prior $p(\theta)$ is **conjugate** to a likelihood $p(\mathcal{D} \mid \theta)$ if the posterior is in the *same parametric family* as the prior.

Likelihood	Parameter	Conjugate Prior	Posterior
Binomial(n, θ)	θ	Beta(α, β)	Beta($\alpha+k, \beta+n-k$)
Poisson(λ)	λ	Gamma(a, b)	Gamma($a+ \sum x_i, b+n$)
Normal(μ, σ^2)	μ	$\mathcal{N}(\mu_0, \tau_0^2)$	$\mathcal{N}(\mu_n, \tau_n^2)$
Normal(μ, σ^2)	σ^2	Inv-Gamma(a, b)	Inv-Gamma(a', b')
Multinomial(n, θ)	θ	Dirichlet(α)	Dirichlet($\alpha + k$)

- Conjugacy yields **closed-form** posteriors — no numerical integration needed
- When no conjugate prior exists, use MCMC (Section 7)

The Beta–Binomial Model

Setting: Flip a (possibly biased) coin n times; observe k heads. Unknown:
 $\theta = \mathbb{P}(\text{head})$.

Model specification

$$k \mid \theta \sim \text{Binomial}(n, \theta), \quad \theta \sim \text{Beta}(\alpha, \beta)$$

Beta density:

$$p(\theta \mid \alpha, \beta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}, \quad \theta \in (0, 1), \quad B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

Key properties:

$$\mathbb{E}[\theta] = \frac{\alpha}{\alpha + \beta}, \quad \text{Mode} = \frac{\alpha - 1}{\alpha + \beta - 2} \quad (\alpha, \beta > 1), \quad \text{Var}[\theta] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Beta–Binomial: Deriving the Posterior

Likelihood: $p(k \mid \theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}$

Prior: $p(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$

Posterior (via proportionality):

$$\begin{aligned} p(\theta \mid k) &\propto p(k \mid \theta) p(\theta) \\ &\propto \theta^k (1 - \theta)^{n-k} \cdot \theta^{\alpha-1} (1 - \theta)^{\beta-1} \\ &= \theta^{(\alpha+k)-1} (1 - \theta)^{(\beta+n-k)-1} \end{aligned}$$

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Conjugacy Result

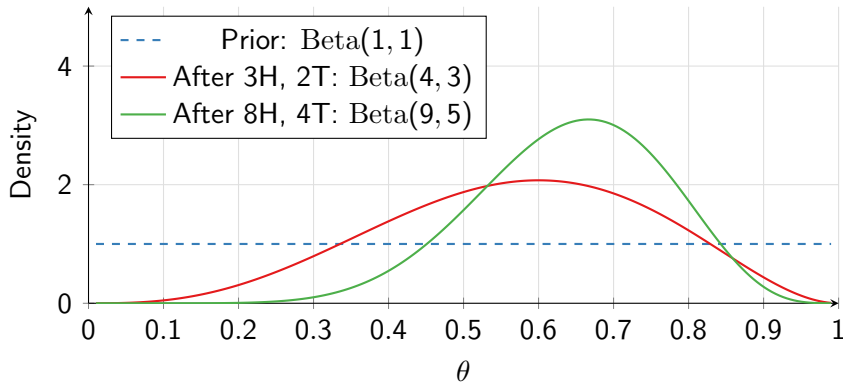
$$\theta | k \sim \text{Beta}(\alpha + k, \beta + n - k)$$

- α and β act as **pseudo-counts**: $\alpha \equiv$ prior successes, $\beta \equiv$ prior failures

- Posterior mean = $\frac{\alpha + k}{\alpha + \beta + n}$ interpolates between prior mean and MLE k/n

Sequential Bayesian Updating

Experiment: Start with $\text{Beta}(1, 1)$ (uniform prior); flip coins and update after each batch.



As data accumulate the posterior concentrates; the flat prior is overwhelmed.

Point Estimates from the Posterior

The posterior $p(\theta \mid \mathcal{D})$ is the complete Bayesian answer. Three common scalar summaries:

Estimator	Definition	Loss minimised	Beta-Binomial
Posterior mean	$\mathbb{E}[\theta \mid \mathcal{D}]$	$(\hat{\theta} - \theta)^2$	$\frac{\alpha + k}{\alpha + \beta + n}$
Posterior median	median of $p(\theta \mid \mathcal{D})$	$ \hat{\theta} - \theta $	$\approx \frac{\alpha + k - \frac{1}{3}}{\alpha + \beta + n - \frac{2}{3}}$
MAP	$\arg \max_{\theta} p(\theta \mid \mathcal{D})$	0-1 loss	$\frac{\alpha + k - 1}{\alpha + \beta + n - 2}$

- Under a flat prior, MAP = MLE; posterior mean $\neq$ MLE
- For asymmetric or multimodal posteriors, *report the full distribution*

Credible Intervals vs. Confidence Intervals

Bayesian Credible Interval

A 95% credible interval $[a, b]$ satisfies:

$$\mathbb{P}(\theta \in [a, b] \mid \mathcal{D}) = 0.95$$

- Direct probability statement about θ
- HPD region: shortest such interval
- θ is random; $[a, b]$ is fixed given $\mathcal{D}$

Frequentist Confidence Interval

A 95% CI is a random interval $[\hat{L}, \hat{U}]$:

$$\mathbb{P}(\theta \in [\hat{L}(\mathcal{D}), \hat{U}(\mathcal{D})]) = 0.95$$

- θ is fixed; the interval is random
- Probability is over repeated samples of $\mathcal{D}$
- *Cannot* say: $\mathbb{P}(\theta \in \text{obs. CI}) = 0.95$

Common Misconception

The frequentist 95% CI **does not** mean there is a 95% probability that θ lies in the observed interval — the credible interval is what most researchers intuitively want.

The Normal–Normal Model

Setting: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, known σ^2 ; infer μ .

Model

$$\mu \sim \mathcal{N}(\mu_0, \tau_0^2), \quad X_i \mid \mu \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$$

Posterior (conjugate update):

$$\mu \mid \mathbf{X} \sim \mathcal{N}(\mu_n, \tau_n^2),$$
$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \quad \mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{X}}{\sigma^2} \right)$$

- Posterior *precision* = prior precision + data precision
- μ_n is a **precision-weighted** average of μ_0 and $\bar{X}$
- As $n \rightarrow \infty$: $\mu_n \rightarrow \bar{X}$ — data dominate the prior
- See Appendix for derivation via completing the square

Hierarchical (Multilevel) Models

Motivation: Observations arise in groups (schools, patients, cities). Treat group parameters as exchangeable draws from a shared hyperprior.

Two-level hierarchy

$$\phi \sim p(\phi) \quad (\text{hyperprior})$$

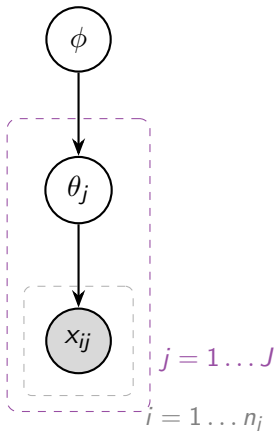
$$\theta_j \mid \phi \sim p(\theta \mid \phi) \quad (\text{group prior})$$

$$x_{ij} \mid \theta_j \sim p(x \mid \theta_j) \quad (\text{likelihood})$$

for groups $j = 1, \dots, J$, observations
 $i = 1, \dots, n_j$.

- **Partial pooling:** extreme group estimates are shrunk toward the grand mean
- Borrows statistical strength across groups — crucial when n_j is small

Plate Diagram



The Computational Challenge

Bayes' theorem requires:

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) p(\theta)}{p(\mathcal{D})}, \quad p(\mathcal{D}) = \int_{\Theta} p(\mathcal{D} \mid \theta) p(\theta) d\theta$$

The Intractability Problem

In most realistic models the integral $p(\mathcal{D})$ is **analytically intractable**. The parameter space Θ may have hundreds to millions of dimensions.

Computational strategies:

- 1 **Conjugate priors** — closed-form update (limited model class)
- 2 **Laplace approximation** — approximate posterior as Gaussian
- 3 **Variational Inference (VI)** — optimisation-based approximation
- 4 **Markov Chain Monte Carlo (MCMC)** — asymptotically exact sampling

Markov Chain Monte Carlo

Core idea: Construct a Markov chain whose *stationary distribution* is $\pi(\theta) = p(\theta \mid \mathcal{D})$.

Metropolis–Hastings Algorithm

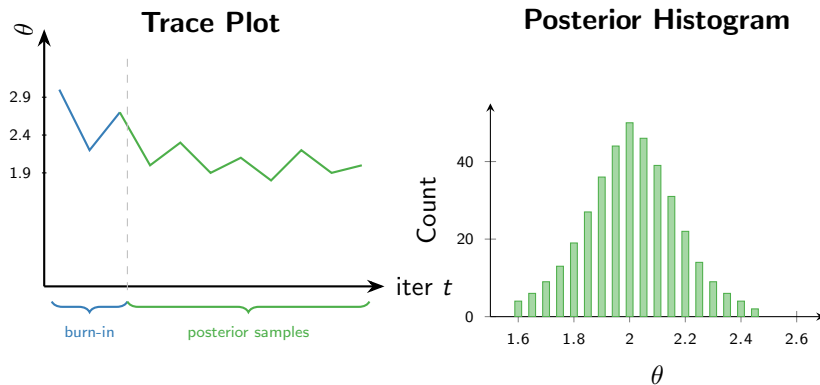
Given current state $\theta^{(t)}$:

- 1 Propose $\theta^* \sim q(\theta^* \mid \theta^{(t)})$ (e.g., $\mathcal{N}(\theta^{(t)}, s^2)$)
- 2 Compute acceptance ratio:

$$\alpha = \min \left(1, \frac{p(\mathcal{D} \mid \theta^*) p(\theta^*) q(\theta^{(t)} \mid \theta^*)}{p(\mathcal{D} \mid \theta^{(t)}) p(\theta^{(t)}) q(\theta^* \mid \theta^{(t)})} \right)$$

- 3 Set $\theta^{(t+1)} = \theta^*$ with probability α ; else $\theta^{(t+1)} = \theta^{(t)}$
- $p(\mathcal{D})$ cancels in the ratio — never needed explicitly!
 - Ergodic chains converge to $\pi(\theta)$ from any starting point

MCMC in Pictures



Discard burn-in samples; the remaining $\{\theta^{(t)}\}$ approximate $p(\theta \mid \mathcal{D})$. Posterior mean $\approx \frac{1}{T} \sum_t \theta^{(t)}$.

Modern MCMC: Hamiltonian Monte Carlo

Problem with MH: random-walk proposals are inefficient in high dimensions.

HMC Key Idea (Neal, 2011)

Augment θ with auxiliary momentum $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$:

$$H(\theta, \mathbf{r}) = \underbrace{-\log p(\theta \mid \mathcal{D})}_{\text{potential energy}} + \underbrace{\frac{1}{2} \mathbf{r}^\top \mathbf{M}^{-1} \mathbf{r}}_{\text{kinetic energy}}$$

Simulate Hamiltonian dynamics via *leapfrog* integration (L steps, size ε):

$$\mathbf{r}_{t+\varepsilon/2} = \mathbf{r}_t - \frac{\varepsilon}{2} \nabla_{\theta} U(\theta_t), \quad \theta_{t+\varepsilon} = \theta_t + \varepsilon \mathbf{M}^{-1} \mathbf{r}_{t+\varepsilon/2}, \quad \mathbf{r}_{t+\varepsilon} = \mathbf{r}_{t+\varepsilon/2} - \frac{\varepsilon}{2} \nabla_{\theta} U(\theta_{t+\varepsilon})$$

- Uses gradient $\nabla_{\theta} \log p(\theta \mid \mathcal{D})$ for directed, long-range moves
- Scales well to hundreds or thousands of parameters
- **NUTS** (No-U-Turn Sampler) automates ε and L — used in Stan, PyMC, Turing.jl

Bayesian Model Comparison

Goal: choose between competing models $\mathcal{M}_1, \mathcal{M}_2$.

Bayes Factor

$$\text{BF}_{12} = \frac{p(\mathcal{D} \mid \mathcal{M}_1)}{p(\mathcal{D} \mid \mathcal{M}_2)} = \frac{\int p(\mathcal{D} \mid \theta_1, \mathcal{M}_1) p(\theta_1 \mid \mathcal{M}_1) d\theta_1}{\int p(\mathcal{D} \mid \theta_2, \mathcal{M}_2) p(\theta_2 \mid \mathcal{M}_2) d\theta_2}$$

$\log_{10} \text{BF}_{12}$	Evidence for $\mathcal{M}_1$
0 – 0.5	Barely worth mentioning
0.5 – 1	Substantial
1 – 2	Strong
> 2	Decisive

- The marginal likelihood $p(\mathcal{D} \mid \mathcal{M})$ automatically penalises model complexity (Occam's razor)

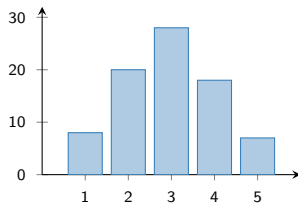
Posterior Predictive Checking

Goal: Does the fitted model generate data similar to observations?

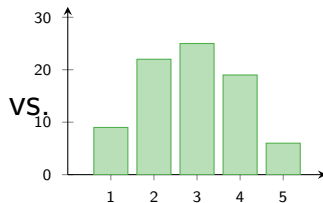
Posterior Predictive Distribution

$$p(\tilde{x} \mid \mathcal{D}) = \int p(\tilde{x} \mid \theta) p(\theta \mid \mathcal{D}) d\theta \approx \frac{1}{T} \sum_{t=1}^T p(\tilde{x} \mid \theta^{(t)})$$

Observed $\mathcal{D}$



Replicated $\tilde{\mathcal{D}}$

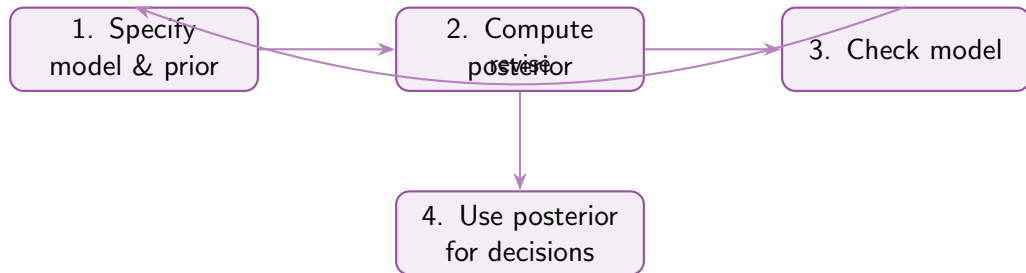


If simulated replicates look unlike $\mathcal{D}$, the model is *misspecified* — revise and iterate.

Key Takeaways

- 1 **Bayes' theorem** turns a prior and likelihood into a posterior:
$$p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta) p(\theta)$$
- 2 The **posterior** is the complete answer; point estimates and intervals are derived summaries.
- 3 **Conjugate priors** give closed-form posteriors for exponential-family likelihoods.
- 4 **Credible intervals** make direct probability statements about θ ; frequentist confidence intervals do not.
- 5 **Hierarchical models** pool information across groups via shared hyperpriors.
- 6 **MCMC** (especially HMC/NUTS) provides scalable sampling from intractable posteriors.
- 7 **Posterior predictive checking** and Bayes factors enable principled model criticism and comparison.

The Bayesian Workflow



Recommended Software

Stan (rstan, cmdstanpy) — HMC/NUTS

PyMC (Python) — probabilistic programming

Turing.jl (Julia) — composable

JAGS / BUGS — Gibbs, classical

Further Reading

Textbooks

- Gelman et al. *Bayesian Data Analysis*, 3rd ed. (BDA3)
- MacKay, *Information Theory, Inference, and Learning Algorithms*
- McElreath, *Statistical Rethinking*
- Bernardo & Smith, *Bayesian Theory*

Foundational Papers

- Bayes (1763) — *Phil. Trans. Roy. Soc.*
- Metropolis et al. (1953) — JACS
- Neal (2011) — *Handbook of MCMC*
- Hoffman & Gelman (2014) — NUTS, *JMLR*

Questions?

Appendix: Normal–Normal Posterior Derivation

Let $X_i \mid \mu \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, $\mu \sim \mathcal{N}(\mu_0, \tau_0^2)$.

Log-posterior:

$$\begin{aligned}\log p(\mu \mid \mathbf{X}) &\propto -\frac{(\mu - \mu_0)^2}{2\tau_0^2} - \sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2} \\ &= -\frac{\mu^2}{2} \underbrace{\left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)}_{1/\tau_n^2} + \mu \underbrace{\left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2}\right)}_{\mu_n/\tau_n^2} + \text{const}\end{aligned}$$

Completing the square yields a quadratic in μ , so:

$$\mu \mid \mathbf{X} \sim \mathcal{N}(\mu_n, \tau_n^2), \quad \frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \quad \mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2} \right)$$

As a convex combination: $\mu_n = (1 - w)\mu_0 + w\bar{x}$, where $w = \frac{n/\sigma^2}{1/\tau_0^2 + n/\sigma^2}$.

Appendix: Gibbs Sampling

Setting: joint posterior $p(\theta_1, \theta_2 \mid \mathcal{D})$; full conditionals tractable.

Gibbs Sampler

Initialise $\theta_1^{(0)}, \theta_2^{(0)}$. For $t = 1, 2, \dots, T$:

- 1 Draw $\theta_1^{(t)} \sim p(\theta_1 \mid \theta_2^{(t-1)}, \mathcal{D})$
- 2 Draw $\theta_2^{(t)} \sim p(\theta_2 \mid \theta_1^{(t)}, \mathcal{D})$

- Each full conditional is a Metropolis–Hastings step with *acceptance probability 1*
- Stationary distribution is the joint $p(\theta_1, \theta_2 \mid \mathcal{D})$
- Easy to implement when full conditionals are in standard families
- Can mix slowly when θ_1 and θ_2 are highly correlated
- Extends naturally to $d > 2$ parameters: cycle through $p(\theta_k \mid \theta_{-k}, \mathcal{D})$ for $k = 1, \dots, d$